Maxime Houlette

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Research & Technical Skills

Numerical Methods: Finite Element Methods, Lattice Boltzmann Methods, Spectral and Discontinuous Galerkin Methods, Krylov Methods, KNN

Mathematics: Functional Analysis, Numerical Analysis, Linear & Non-Linear Approximation, Incompressible Fluid Mechanics, Structural Mechanics

Programming: Python, C++, LaTeX, Git/GitLab

Languages: French (native), English (B2), Japanese (A2)

Research Experience

- **Inria — Lille, France** *April 2026 – Present*
  Research Intern
  - Analysis of a Discontinuous Galerkin (DG) scheme for the Helmholtz equation in the high-frequency regime
  - Derivation and improvement of asymptotic mesh conditions on the polynomial degree and mesh size guaranteeing quasi-optimality of the discrete solution
  - Implementation and numerical validation in C++; version control via GitLab

- **The University of British Columbia — Vancouver, Canada** *January 2025 – July 2025*
  Visiting International Research Student
  - Development of a *Distributed Lagrange Multipliers / Fictitious Domain* method for fluid–rigid body interactions
  - Coupling of the DLMFD method with a monophasic *Direction Splitting* Navier-Stokes solver
  - Implementation in C++; version control via GitLab

- **DotBlocks — Station F, Paris, France** *June 2024 – December 2024*
  Research Intern
  - Design and implementation of a Lattice Boltzmann scheme for the simulation of compressible and incompressible Navier-Stokes equations
  - Analysis of boundary treatment strategies for simple and curved 2D obstacles
  - Implementation in Python; version control via GitLab

Education

- **Sorbonne Université — Paris, France** *September 2025 – Present*
  Mathematical Modelling — Master 2
  - Functional & Numerical Analysis, Finite Element Methods, Krylov Methods, Linear and Non-Linear Approximation, Spectral Methods, Discontinuous Galerkin Methods

- **École nationale des ponts et chaussées, Institut Polytechnique de Paris — Paris, France** *2022 – Present*
  Génie Mécanique des Matériaux — Master 1
  - Physical Mechanics of Materials, Structural Mechanics, Incompressible Fluid Mechanics, Finite Element Theory

- **Lycée Malherbe — Caen, France** *2020 – 2022*
  - *Classes Préparatoires aux Grandes Écoles* (MPSI–MP*)

- **Lycée Charles de Gaulle — Caen, France** *2020*
  - Scientific *Baccalauréat*, with highest honors (*mention très bien*)

Additional Experience

- **Private Tutor in Mathematics, Physics and Chemistry** *October 2022 – Present*
  - Weekly tutoring sessions for high school and *classe prépa* students

- **School Bar Association — École nationale des ponts et chaussées** *2023 – 2024*
  - Stock management and event organisation within a 15-member team (two events a week)

Interests

Football (vice-captain of the university team), tennis, skiing; self-taught guitar and piano